

Line Number	Identifier	Error Type	Error	Error Correction
1	Header File (<stdio.h>)			
2	Header File (<string.h>)			
4	Function Definition (main)			
6	Array Declaration (int marks[10])			
8	Function Call (printf)			
10	Variable Declaration (int total)			
12	Loop Declaration (for)			
14	Assignment (total += marks[i])			
17	Function Call (printf)			
19	Variable Declaration (float average)			
21	Assignment (average = total / 10)	Logical Error	Integer division occurs before assignment to float	average = total / 10.0;
23	Function Call (printf)	Semantic Error	%d format specifier used with float	printf("Average = %f\n", average);
27	Variable Declaration (int highest)			
29	Loop Declaration (for)			
31	Conditional Statement (if)	Logical Error	Condition identifies lowest value instead of highest	if(marks[i] > highest)
33	Assignment (highest = marks[i])			
39	Variable Declaration (int lowest)			
41	Loop Declaration (for)			
43	Conditional Statement (if)	Logical Error	Condition identifies highest value instead of lowest	if(marks[i] < lowest)
45	Assignment (lowest = marks[i])			
51	Array Declaration (int attendance[10])			
53	Loop Declaration (for)			
55	Array Element Assignment (attendance[i])			
58	Array Access (attendance[10])	Runtime Error	Index 10 exceeds valid range 0-9	Use indices 0-9 only
60	Variable Declaration (int present)			
62	Loop Declaration (for)	Runtime Error	Loop condition i<=10 accesses attendance[10]	for(int i=0;i<10;i++)
64	Assignment (present += attendance[i])			
67	Function Call (printf)			
71	Array Declaration (char subject[10])			
73	Function Call (strcpy)	Runtime Error	Destination array too small for source string "DataStructures"	Increase array size
75	Function Call (printf)			
79	Array Declaration (char department[20])			
81	String Assignment (department = "InformationTechnology")	Semantic Error	Character arrays cannot be assigned directly	strcpy(department, "InformationTechnology");
83	Function Call (printf)			
87	Array Declaration (int fees[5])			
89	Array Access (fees["two"])	Semantic Error	String literal used as array index	Use integer index
91	Array Access (fees[4.5])	Semantic Error	Floating-point value used as array index	Use integer index
93	Loop Declaration (for)			
95	Array Element Assignment (fees[i])			
98	Function Call (printf)			
100	Loop Declaration (for)			
102	Function Call (printf)			
107	Array Declaration (int result[5])			
109	Loop Declaration (for)			
111	Function Call (scanf)	Semantic Error	Array element passed without address operator	scanf("%d", &result[i]);
116	Variable Declaration (int totalResult)			
118	Loop Declaration (for)			

120	Assignment (totalResult += result[i])			
123	Function Call (printf)			
127	Array Declaration (int inventory[5])			
129	Conditional Statement (if)	Logical Error	Assignment operator used instead of comparison	if(inventory[2] == 30)
131	Function Call (printf)			
136	Array Declaration (int backup[5])			
138	Array Assignment (backup = inventory)	Semantic Error	Arrays cannot be assigned directly	Use memcpy() or copy elements individually
140	Function Call (printf)			
144	Array Declaration (float salary[5])			
146	Array Element Assignment (salary[0])			
148	Function Call (printf)	Semantic Error	%d format specifier used with float	printf("%f\n", salary[0]);
152	Array Declaration (char city[10])			
154	String Assignment (city = "Jaipur")	Semantic Error	Character arrays cannot be assigned directly	strcpy(city, "Jaipur");
156	Function Call (printf)			
160	Array Declaration (char branch[5])			
162	Array Element Assignment (branch[0])			
163	Array Element Assignment (branch[1])			
164	Array Element Assignment (branch[2])			
165	Array Element Assignment (branch[3])			
167	Function Call (printf)	Runtime Warning	Character array is not null-terminated before printing with %s	Add branch[4] = '\0';
171	Array Declaration (int stock[5])			
173	Loop Declaration (for)	Runtime Error	Loop condition i<=5 accesses stock[5]	for(int i=0;i<5;i++)
175	Array Element Assignment (stock[i])			
178	Function Call (printf)			
180	Loop Declaration (for)			
182	Function Call (printf)			
187	Array Declaration (char course[8])			
189	Function Call (strcpy)	Runtime Error	Destination array too small for source string "Programming"	Increase array size
191	Function Call (printf)			
195	Array Declaration (int ratings[5])			
197	Array Access (ratings[-1])	Runtime Error	Negative index access is invalid	Use indices 0-4
199	Function Call (printf)			
203	Array Declaration (int feedback[5])			
205	Function Call (printf)	Semantic Error	%s format specifier used with integer argument	printf("%d\n", feedback[0]);
209	Array Declaration (int scores[5])			
211	Array Assignment (scores = marks)	Semantic Error	Arrays cannot be assigned directly	Use memcpy() or copy elements individually
213	Function Call (printf)			
217	Array Declaration (int temp[5])			
219	Loop Declaration (for)			
221	Array Element Assignment (temp[i])			
224	Function Call (printf)	Runtime Error	temp[5] is outside valid range 0-4	Use temp[0] to temp[4]
228	Function Call (printf)			
230	Return Statement			